

Analytical evaluation of the interdigital electrodes capacitance for a multi-layered structure

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Abstract

The widespread use of interdigital electrodes in such applications as microwave filters, surface acoustic wave devices, electro-optic shutters as well as on chemical and biological sensing and even on the electrical and dielectric characterization of materials requires that we improve our description of their electrical performance.

In this paper, we present new analytical expressions for the capacitance between the two comb electrodes of a periodic interdigital capacitive sensor, based on conformal mapping techniques. This proposed model is general and quite independent of the particular application and can be applied for any space and finger width as well as for any number of layers with different thickness and permittivity. The capacitance for a particular sensor configuration is a function of the dielectric permittivity of the materials, the fingers length and of two geometric non-dimensional parameters: (i) the ratio between the space and finger widths; (ii) the ratio between the thickness of the sensitive layer and the spatial sensor wavelength. Comparisons with previously published models as well as with experimental data and finite element analysis were made.

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1. Introduction

Interdigital capacitors (IDC) for technological applications have been studied by many authors since the early 1970s. The applications for these structures include their use in lumped elements for microwave integrated circuits [1,2], optical and surface acoustic wave devices [3], optically controlled microwave devices [4], thin-film acoustic–electronic transducers and tunable devices [5], dielectric studies on thin films [6]. More recently, studies had been made with IDC for humidity [7,8] and chemical sensors [9–12]. Typical interdigital capacitive sensors (IDC-S), are made of an ‘inert’ substrate over which the two comb electrodes are deposited. One chemical sensitive layer (usually a polymer) is then deposited over the electrodes. Polymers have been used for organic vapor sensing because they exhibit rapid reversible vapor sorption and are easy to apply as thin or thick films by a variety of techniques [13]. The polymer layer can be chosen according to its affinity to a particular molecule or

set of molecules one wishes to detect. If several sensors with different polymer layers are used to make a sensor array it is then possible to evaluate complex organic vapor samples [14]. These sensor arrays can be part of an electronic nose for a particular application depending on the sensitive layers that are chosen.

We are developing a multi-parameter sensor array, made with IDC-S to be part of an artificial olfactory system for an on-line and real-time monitoring of the aroma profile during the muscatel wine-fermentation. The choice of these sensors enable a wide choice of polymeric sensitive layers, especially oriented for this case studied [15,16]. The change in capacitance of these sensors depends on the changes in the dielectric permittivity and on swelling of the polymer (sensitive) layer due to the ad/absorption of the analyte.

The correct design of an interdigital capacitive sensor (IDC-S) requires closed form expressions for the computation of capacitance, based on the geometry of the sensor and the properties of the substrate and chemical sensitive layer(s).

One of the early models for the interdigital capacitor was published in 1970 by Alley [1] and was based on a lossless coupled microstrip line theory. This is an approximate

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model, which can predict the capacitance values for equal (or closed, based on an error function) finger and gap width, and with an infinite top air layer. This model is still used, since experimental values are in good agreement with the proposed method of capacitance computation [17,18].

Alley's model had been modified by Hobdell [19] in 1979 introducing a loss term into Alley's theory, and by Esfandiari et al. [20] in 1983, which incorporates the influence of the metallization thickness into the total value of the capacitance.

In 1977 Wei [3] proposed a model based on conformal mapping (CM) techniques to evaluate the capacitance of an IDC with an infinite top air layer.

The restrictions imposed in these models are often not adequate for the evaluation of IDC-S, since they cannot predict the capacitance for a sensor having a finite layer or a multi-layer structure over the electrodes.

The first model taking into account for such multi-layered structure was proposed by Wu et al. [5] in 1994. This model is based on the work of Veyres and Hanna [21] giving rise to higher values of capacity than that we find experimentally.

In 1996 Gevorgian et al. [22] proposed a different model for IDC that could take into account a multi-layered top structure. This model was again based on CM techniques and differs from Wu model on the choice of the unit cell as a repeated unit, as well it accounts for the error introduced by the finger endings and sensor boundaries. Although the values computed with Gevorgian model they do not agree neither with experimental values nor finite element analysis.

In the present work, original conformal transformations and the partial capacitance method [2] are used to evaluate closed form expressions for computation of capacitances of IDC on N-layered substrates, taking into account the capacitances between the fingers as well as the fringing capacitance in the outer fingers. With this model, we obtain new and more exact expressions for the calculation of the capacitance for a general multi-layered structure, which are corroborate with experimental values for the particular case of an infinite top air layer and with finite element analysis for the case of a finite top layer. As well for the first time a consistent complete model is present which can be confirmed by noticing that the expressions for the particular case of an infinite top layer can be obtained from the more general case of a finite top layer making its height approach infinite.

2. Theory

2.1. Physical model of IDC-S

Fig. 1 shows a layout of the electrode plane and a schematic diagram of the cross-section of an interdigital capacitor sensor (IDC-S) with two interpenetrating comb electrodes. The gaps between electrodes have a width of G while the fingers have a width of W . Each comb electrode

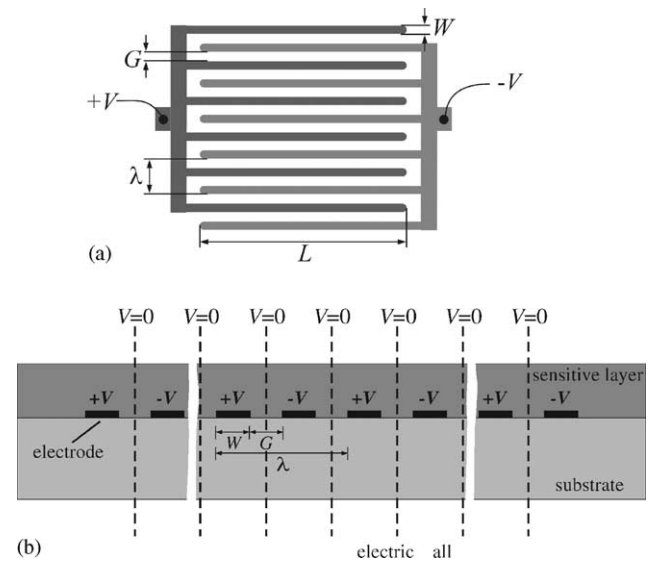


Fig. 1. (a) Layout of electrode plane; (b) cross-section of a periodic IDC-S showing the electric potential boundary planes distribution.

is connected to a fixed potential (either $+V$ or $-V$) and has a number of fingers of length L . The electrodes, made of gold in general, are deposited on a substrate whose thickness is much greater than the IDC-S spatial wavelength ($\lambda = 2(W + G)$). A sensitive layer is deposited over the electrodes. The capacitance of the IDC-S only changes as long as there is also a change in the capacitive properties of the sensitive layer or in its thickness. The metallization ratio is thus

$$\eta = \frac{W}{W + G} = \frac{2W}{\lambda} \quad (1)$$

This adimensional parameter, η (e.g. $\eta = 0.5$ means 50% of metallization), has great relevance, since the capacitance strongly depends on this parameter, as will be shown in the next sections.

By symmetry, the perpendicular planes halfway between the electrodes are equipotential planes with $V = 0$ (ground electric walls), with electric field lines crossing normal to these equipotential planes. This holds exactly true for an infinite periodic structure obeying the Laplace equation (i.e. with no electric charges present) and whose electrodes fingers have an infinite length.

In the case of a finite periodic structure there is an error incurred by the previous assumption due to a bending of the ground walls, which is especially true, closer to the outer electrodes. In this work it is assumed, nevertheless, that most of the polarization phenomena occurs in a region where the bending of the ground planes is not significant.

In relation to the finite length of the electrodes strips, they are much greater than the wavelength of the IDC-S structure so that for practical purposes they can be considered infinite.

One can also neglect the thickness of the electrodes (~ 50 nm) since they are much thinner than their width ($> 50 \mu\text{m}$), and thus, the electric potential of the electrodes

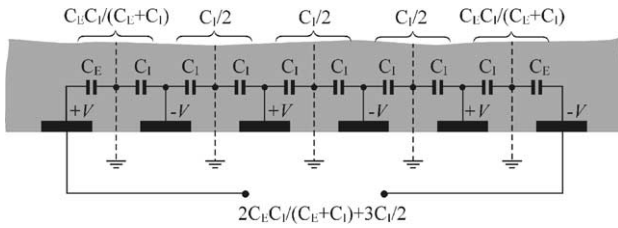


Fig. 2. Example of the equivalent circuit for evaluation of the static capacitance of a semi-infinite top layer in a periodic IDC-S with six electrodes.

is specified at the interface between the upper and lower half planes. For thicker electrodes corrections can easily be made to take into account their thickness [2,22].

2.2. Single semi-infinite layer capacitance

We first calculate the capacitance of a semi-infinite layer above the electrode plane using an equivalent electric circuit. In Section 2.3, the partial capacitance technique is introduced to account for a multi-layer structure above the electrode plane.

For symmetry reasons, the capacitance of one single semi-infinite layer can be evaluated as a function of two types of capacitance (see Fig. 2): (1) C_I —being half the capacitance of one interior electrode relative to the ground potential and (2) C_E —the capacitance of one outer electrode relative to the ground plane next to it. Fig. 2 shows the equivalent electric circuit for a structure of six electrodes having a semi-infinite layer above the electrode plane.

Using network analysis to evaluate the equivalent circuit of Fig. 2 one finds the total capacitance between the negative and positive electrodes of a semi-infinite layer IDC-S to be equal to

$$C = (N - 3) \frac{C_I}{2} + 2 \frac{C_I C_E}{C_I + C_E}, \quad N > 3 \quad (2)$$

where N is the number of electrodes. Structures with $N = 2$ and 3 cannot be considered in the scope of this work, although these structures can be easily studied with small modifications of the present model. There are available nevertheless, closed form solutions for these two configurations [2,21].

The capacitance for the lower half plane is calculated in the same way and adds to the capacitance calculated for the upper half plane to obtain the total capacitance of the sensor.

2.3. N-layer capacitance

The capacitances C_I and C_E can also be calculated for a sensor having one or more finite dielectric layers (of different permittivities and thickness) on the top of the electrodes using the so-called partial capacitance technique. Despite the validity of this technique was not already completed demonstrated it have been widely used by several authors given accurate results [21,23–25]. A simple representation of the splitting of a two-layered sensor (in the upper half plane) is shown in Fig. 3, where ϵ_1 and ϵ_2 are the relative dielectric constants of layers 1 and 2, respectively. The capacitance of a sensor taking into account the different layers is the sum of the partial capacitances as follows:

$$C_u = C_{h=\infty} + (\epsilon_1 - 1)C_{h=h_1} + (\epsilon_2 - \epsilon_1)C_{h=h_2} \quad (3)$$

where C_u is the total capacitance of the upper half plane and C_h is the geometric capacitance of one layer, which depends on its height h and on the particular electrode geometry.

In the present work two types of electrode geometry (i.e. interior and exterior electrodes) are considered and one has therefore, to compute both a $C_{I,u}$ and a $C_{E,u}$ which stand for the multi-layer interior and exterior electrode capacitances, respectively. The total upper capacitance of the sensor is then calculated as previously described using Eq. (2).

2.4. Computation of an interior half-capacitance C_I

We will use conformal mapping techniques to map an appropriate space region of the IDC-S onto a parallel plate capacitor geometry for which the capacitance value is known as well as the equipotential lines and the lines of the electric field. As the conformal transformation is analytic can be shown that the capacitance in the original space region will transform accordingly [26].

We now proceed to obtain the expression for half the capacitance between an interior electrode and the ground potential, with a dielectric layer of thickness h over the electrodes. The first step is the definition of a physical domain in the Argand complex x -plane (see Fig. 4) whose horizontal boundaries coincide with those of the dielectric layer while the vertical limits are (a) the perpendicular line at half electrode denoted as $x_1 = 0$ and (b) the vertical line passing by $x_3 = \lambda/4$ which, by symmetry, constitutes an equipotential line and thus an electric wall. The shadowed regions in Fig. 4 represent the dielectric layer with thickness h .

This rectangle on the x -plane is first mapped onto the complex z -plane keeping the same aspect ratio but written in



Fig. 3. Splitting of a two-layered half plane according to the partial capacitance technique.

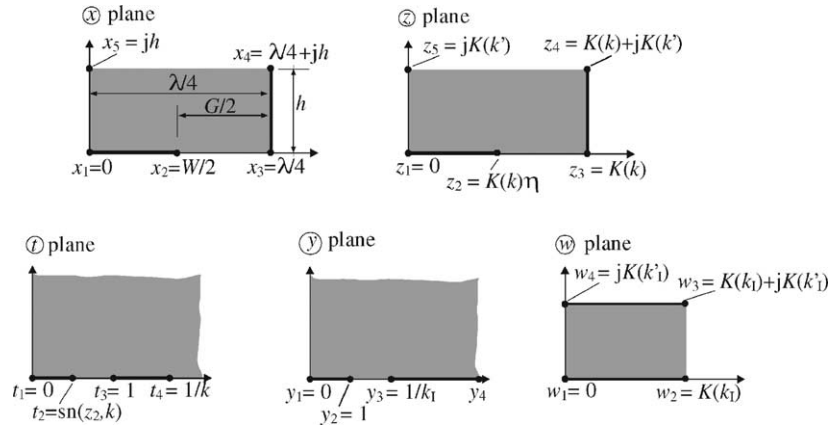


Fig. 4. Conformal transformations for the calculation of C_1 . The fixed equipotential lines and their transformations are marked as solid lines while the dielectric and its transformations are the shadowed regions.

terms of the elliptic integral of first kind. The transformation should be in agreement with the following module equation:

$$4r = \frac{K(k')}{K(k)} \quad (4)$$

where $K(k)$ is the complete elliptic integral of first kind with modulus k , k' the complementary modulus, $k' = \sqrt{1-k^2}$, and

$$r = \frac{h}{\lambda} \quad (5)$$

This adimensional parameter, r , is again of great relevance since together with the parameter η it will determine the capacitance of a particular sensor as we would see in Sections 2.5 and 2.6.

The value of the modulus k that satisfies the module equation (4) is calculated as

$$k = \left(\frac{v_2(0, q)}{v_3(0, q)} \right)^2 \quad (6)$$

where v_2 and v_3 are the Jacobi theta functions [27] with

$$q = \exp\left(-\pi \frac{K(k')}{K(k)}\right) = \exp(-4\pi r) \quad (7)$$

Thus, the mapping function from the x -plane into the z -plane is

$$z = \frac{4K(k)}{\lambda} x \quad (8)$$

The rectangle in the z -plane is then transformed into the first quadrant in the t -plane by the function:

$$t = \text{sn}(z, k) \quad (9)$$

where $\text{sn}(z, k)$ is the Jacobi elliptic function of modulus k (see Fig. 4). In the third transformation the first quadrant in the t -plane is transformed onto the first quadrant in the y -plane by means of the function (which extends

the electrode defined by the points t_3 – t_4 on t -plane into a semi-infinite electrode y_3 – y_4 on y -plane):

$$y = \frac{t}{t_2} \sqrt{\frac{t_4^2 - t_2^2}{t_4^2 - t^2}} \quad (10)$$

and finally the y -plane is transformed into the w -plane using the Schwarz–Christoffel transformation [28] that maps the represented quadrant of the y -plane into a rectangle on the w -plane,

$$w = \int_0^y \frac{dw'}{\sqrt{(1-w'^2)(1-k_1^2 w'^2)}} \quad (11)$$

where

$$k_1 = t_2 \sqrt{\frac{t_4^2 - 1}{t_4^2 - t_2^2}} \quad (12)$$

We have thus mapped the rectangular region in the x -plane onto a rectangular region onto the w -plane. The particular feature of this set of transformations is that the electrodes in the x -plane are transformed, in the w -plane, into electrodes of a parallel plate capacitor. As one knows the dimensions of the transformed parallel plate capacitor in the w -plane the capacitance in x -plane can be directly obtained as

$$C_1 = \varepsilon_0 \varepsilon_r L \frac{K(k_1)}{K(k'_1)} \quad (13)$$

where ε_r is the relative permittivity of the dielectric layer and L is the length of the electrodes strips. In Appendix B are tabulated the relevant expressions for the computation of the capacitance C_1 for a finite layer. It can be seen that this capacitance is a function of η , r , ε_r and L (i.e., $C_1 = C_1(\eta, r, \varepsilon_r, L)$) so this capacitance is not dependent of the absolute dimensions of the IDC-S cross-section, since η and r are adimensional.

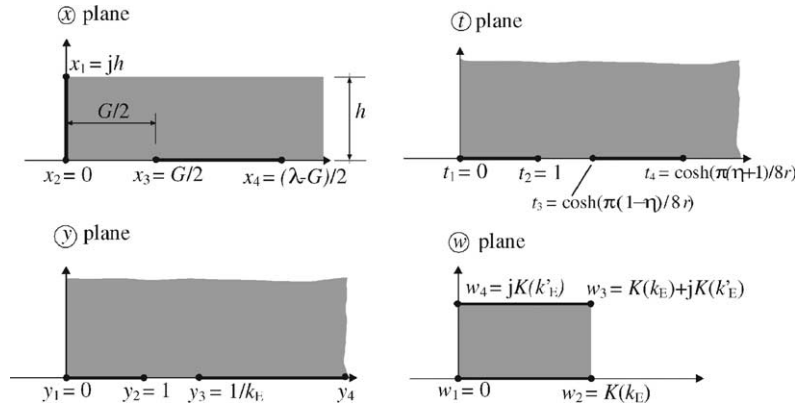


Fig. 5. Conformal transformations for the calculation of C_E . The fixed equipotential lines and their transformations are marked as solid lines while the dielectric and its transformations are the shadowed regions.

2.5. Computation of an exterior half-capacitance C_E

For the calculation of C_E the starting physical domain instead of a rectangle will have the right-hand side extended to infinite (see Fig. 5). This semi-infinite strip will be mapped onto the t -plane by means of the function:

$$t = \cosh\left(\frac{\pi}{2h}x\right) \quad (14)$$

Next, the t -plane is mapped on the y -plane using the mapping function:

$$y = t \sqrt{\frac{t_4^2 - 1}{t_4^2 - t^2}} \quad (15)$$

Finally the y -plane is transformed on the w -plane using the Schwarz-Christoffel transformation that maps the represented quadrant of the y -plane into a rectangle in the w -plane:

$$w = \int_0^y \frac{dw'}{\sqrt{(1-w'^2)(1-k_E^2 w'^2)}} \quad (16)$$

where

$$k_E = \frac{1}{t_3} \sqrt{\frac{t_4^2 - t_3^2}{t_4^2 - 1}} \quad (17)$$

We have thus mapped the semi-infinite strip region in the x -plane onto a rectangular region in the w -plane. Again the electrodes in the x -plane are transformed, in the w -plane, into electrodes of a parallel plate capacitor. Using the dimensions of the transformed parallel plate capacitor in the w -plane the capacitance in x -plane can be directly obtained as

$$C_E = \varepsilon_0 \varepsilon_r L \frac{K(k_E)}{K(k'_E)} \quad (18)$$

In Appendix B are also tabulated expressions relevant for the computation of the capacitance C_E for a finite layer. It can be seen that this capacitance is again function of η , ε_r and L . Therefore, C_E , like C_I , is not dependent on the

absolute dimensions of the IDC-S cross-section but rather on its aspect ratio.

2.6. Limiting cases (infinite height layer)

For any IDC-S, even if it only has one sensitive top layer and according to the partial capacitance technique, is necessary to evaluate the capacitance of an infinite air layer. This means that in the equations for the capacitance, $h \rightarrow \infty$ ($r \rightarrow \infty$). Of course, this can be done by the calculation of the limits of k_I and k_E as the thickness goes to infinite. This analytical approach is not straightforward, and several transformations are needed. Another way is to perform the calculation of the capacitance for an infinite layer using conformal mapping techniques, as is shown and demonstrated in Appendix A, and the final equations are

$$\lim_{h \rightarrow \infty} k_I = k_{I\infty} = \sin\left(\frac{\pi}{2}\eta\right) \quad (19)$$

$$\lim_{h \rightarrow \infty} k_E = k_{E\infty} = \frac{2\sqrt{\eta}}{1+\eta} \quad (20)$$

These expressions show that the $C_{I\infty}$ and $C_{E\infty}$ (capacitances for an infinite layer), like C_I and C_E , do only depend on the metallization ratio η , the r parameter (having now a fixed value ($r \rightarrow \infty$)), ε_r and L .

2.7. Total capacitance

Using the above equations and the partial capacitance technique, a typical IDC-S with a thick substrate of relative permittivity ε_S and one sensitive layer of relative permittivity ε_I , will have their total capacitances $C_{I,IDC}$ and $C_{E,IDC}$ given by

$$\begin{aligned} C_{I,IDC} &= C_{I,air} + C_{I,1} + C_{I,S} \\ &= \varepsilon_0 L \left(\frac{K(k_{I\infty})}{K(k'_{I\infty})} + (\varepsilon_I - 1) \frac{K(k_{I,1})}{K(k'_{I,1})} + \varepsilon_S \frac{K(k_{I\infty})}{K(k'_{I\infty})} \right) \end{aligned} \quad (21)$$

$$C_{E,IDC} = C_{E,air} + C_{E,1} + C_{E,S}$$

$$= \varepsilon_0 L \left(\frac{K(k_{E\infty})}{K(k'_{E\infty})} + (\varepsilon_1 - 1) \frac{K(k_{E,1})}{K(k'_{E,1})} + \varepsilon_S \frac{K(k_{E\infty})}{K(k'_{E\infty})} \right) \quad (22)$$

where the first two terms represent the contribution of the upper half plane while the last term represents the contribution of the lower half plane to the respective capacitances.

The above equations are written for the case of a thick substrate (the thickness of the substrate much higher than $\lambda/2$, see Section 3). This is almost always the case, since it is supposed to be an inert substrate. Nevertheless for other applications the equations can be modified to take into account a thinner substrate, where the substrate acts like a single finite layer having, for example, an infinite air layer below it.

Finally the total capacitance of the IDC-S is given using Eq. (2) as

$$C_{IDC} = (N - 3) \frac{C_{I,IDC}}{2} + 2 \frac{C_{I,IDC} C_{E,IDC}}{C_{I,IDC} + C_{E,IDC}} \quad (23)$$

The above example serves as an indication on how to calculate the total capacitance for an IDC-S sensor having one sensitive layer and a thick substrate. For an IDC-S with more than one top layer and/or for thinner substrates the above equations (Eqs. (21)–(23)) can be easily and suitably modified.

3. Results and discussion

3.1. IDC-S with top air layer

Despite the fact that a typical IDC-S has usually one layer on the top of the electrodes, it is of scientific interest to compare the results obtained with this model to the available models for an IDC with an infinite top air layer, since this configuration is the most studied one. In this case, the expressions for the capacitance computation may be easily deduced from Eqs. (21)–(23) in which $C_{I,1}$ and $C_{E,1}$ are null. In Fig. 6, the dependence of the capacitance on the metallization ratio η is shown and compared to different available models and experimental values.

Experimental data for IDCs were obtained using borosilicate glass as substrate ($\varepsilon_S = 5.4$, $h_S = 2$ mm) with the following geometric parameters: $\lambda = 800$ μ m and W changing from 120 to 280 μ m ($0.3 < \eta < 0.7$). Each capacitor has $N = 54$ electrodes with a length of $L = 2$ cm. A thin film (~ 50 nm) of Cr/Au was deposited on glass substrate by electron-gun and the electrode pattern made by a typical lift-off process. The capacitance was measured with the IDCs under vacuum (10^{-2} mbar) at a temperature of 25 °C and with a frequency of 1 kHz.

The capacitance according to Alley's [1] model for the configuration with $\eta = 0.5$ is 27.4 pF which is close to the value of 30.4 pF computed with the present model. As

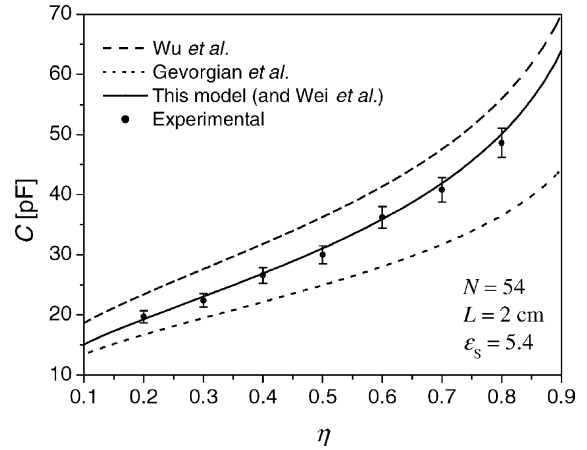


Fig. 6. Capacitance as a function of the parameter η for with an infinite air layer above the substrate; comparison between available models and experimental values. Simulation parameters are written in graph.

mentioned before Alley's model is quite restrictive because it can only be used for a metallization ratio of $\eta = 0.5$ and furthermore, does not enable the use of finite layers over the electrodes. Despite the differences in the mapping functions used by Wei [3], his equation for the capacitance is equivalent to the one of our model for this particular case. In that respect, Wei model gives a better appreciation of the capacitance as a function of the metallization ratio, η , than Alley's equations, but once again cannot account for finite layers on top of the electrodes.

The larger capacitance computed using the equations of Wu et al. [5] is due to the assumptions made on that model. Wu et al. assume that the capacitance of the IDC is a sum of capacitances of unit cells composed of three electrodes, which in fact is not the real case and as a consequence the values computed by Wu et al. are higher than the real capacitance. In fact, a three electrode configuration gives raise to extended electric field lines below and above the electrode plane, when compared with our model. This can be observed using FEM, and will be clear in the next section.

The values for Gevorgian et al. [22] model in Fig. 6 were calculated from their expressions for a finite layer in which the thickness becomes very large when compared to the sensor wavelength. This was done because, the expressions published for that model when $h \rightarrow \infty$, do not give reasonable values. Nevertheless, the values obtained with Gevorgian equations give lower values for the capacitance. We have not found any physical evidence for the lower values obtained by Gevorgian et al., apart from the method used to carry out the conformal transformations which is far too different from the one we use in our model. As will be seen in the next section, even greater differences between our model and the Gevorgian model occurs when one calculates the capacitance of an interdigital capacitor having different layers.

As can be seen in Fig. 6, the experimental values are near the line calculated using our model. The slightly lower val-

ues can be ascribed to the error introduced by the lift-off process that give rise to electrodes that are not perfect lines as the layout mask. On the other hand, since the thickness of electrodes is very small (~ 50 nm) the resistivity of the electrodes are not exactly zero and, this will lower the potential especially near the finger ends.

In order to evaluate the validity of this model for a finite structure with a small number of electrodes, a simulation using FEM was performed in a four-electrode structure. This simulation was made using ANSYS software with the following parameters: ($\eta = 0.5$; $L = 2$ cm; $\varepsilon_s = 3.15$, infinite top air layer). The FEM value obtained for the capacitance was 12.46 pF while, according to our model, this structure has a value of 12.64 pF, which represents an error of 1.4%. This error is of the order of magnitude of the FEM error, giving therefore a clear indication that even for most unfavorable case ($N = 4$) our model gives meaningful values.

3.2. IDC-S with one sensitive layer

This configuration is of much interest for chemical sensing which brings us to the need to understand how the capacitance changes with parameter r , which is the ratio between the height of the sensitive layer and the sensor wavelength. In other words, this will tell us how far from the electrode plane a particular sensor will respond to a change in the permittivity of the sensitive layer. Another aspect to take in account is the possibility of the swell of the sensitive layer, which usually happens in polymeric layers when absorption occurs. This swelling can lead to a measurable change in capacitance if the thickness of the sensitive layer is made to have a value far from its saturation value (see Section 3.3).

In Fig. 7, the dependence of the capacitance on the metallization ratio η for a sensor with one finite layer above the electrodes is shown and compared with the different available models. The simulation parameters are as follows: $\varepsilon_s = 3.15$; $h_s = 2$ mm; $\varepsilon_1 = 5$; $r = 0.25$; $N = 100$; $L = 2$ cm.

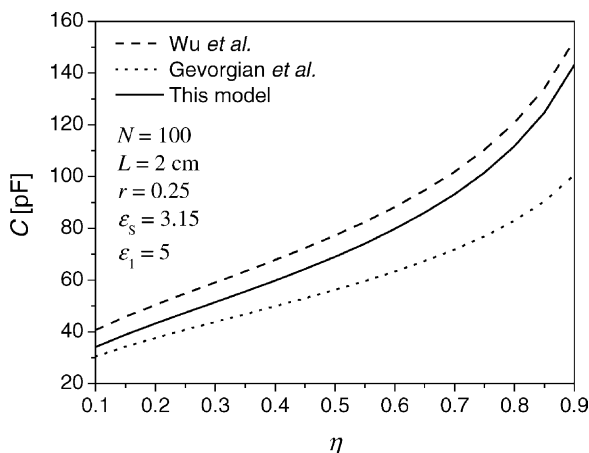


Fig. 7. Capacitance as a function of the metallization ratio, η , for a sensor with one finite sensitive layer above the substrate for different available models. The simulation parameters are written in graph.

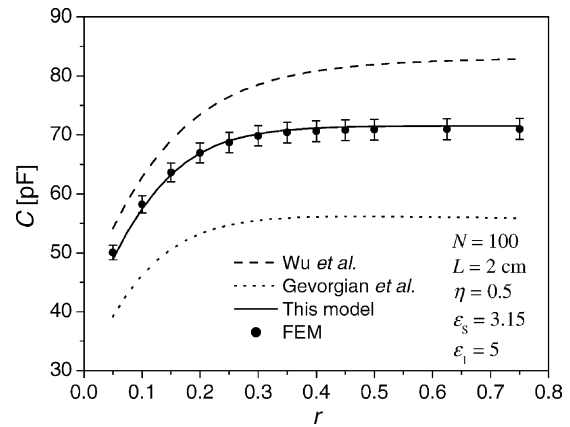


Fig. 8. Capacitance as a function of r ; comparisons between available models and FEM. Simulation parameters are written in graph.

Fig. 8 shows the capacitance as a function of the parameter r for the present model and available models together with FEM. Simulation parameters are: $\eta = 0.5$; $L = 2$ cm; $N = 100$; $\varepsilon_s = 3.15$; $\varepsilon_1 = 5$. The capacitance reaches a saturation value for a value of $r = 0.5$ ($h = \lambda/2$), this means that the sensor is not sensitive to a distance from the electrode plane greater than about half the sensor wavelength. The differences found between our model and FEM can be due to the simulation conditions used in FEM: (i) no corrections for the outer fingers were made in FEM; (ii) the capacitance value is sensitive to the mesh used in the simulation; (iii) with FEM was not possible to extend the horizontal boundaries of the unit cell to infinite. The overall error for the FEM was estimated to be of about 2%. Nevertheless the behavior of the capacitance value as evaluated by FEM is in agreement with our model, reaching the saturation (i.e. 99.5% of $C(h = \infty)$) for the same value of r .

Apart from the large discrepancies in the capacitance values calculated by Wu et al. and Gevorgian et al. as mentioned in the previous section, other significant differences are present. In the Wu model the saturation of the capacitance occurs for a large value of r (i.e. $r = \lambda$), which is consequence of the method of Wu et al., since the capacitance is calculated as if the periodic structure is a combination of unit cells with three electrodes. This will give a different distribution of the electric field lines, with higher values of electric field for a particular distance from the electrodes plane.

Gevorgian model does not present any saturation value for the capacitance, in fact, the capacitance will decrease for larger values of h which has no physical meaning, since the permittivity of the medium is higher than that of air.

3.3. IDC-S design considerations

Some considerations have to be taken into account for a correct design of an IDC-S with a particular substrate and sensitive layer: (i) the spatial sensor wavelength, since

every technique for the drawing of the electrodes has its limitations (e.g. normal low cost photolithographic process has a limitation of gap and electrode width of about 50 μm); (ii) the area of implementation, which depends on the size we want for our sensor; (iii) desirable nominal value of the sensor capacitance; (iv) the expected change on the sensor capacitance.

The nominal value of the sensor capacitance together with its expected change has a great influence on the electronics associated with the signal conditioning of an IDC-S. The area of implementation and the sensor capacitance can be joined together as a new parameter: the capacitance density. When this value is fixed we can find the correspondent η , r pairs, and consequently the best sensor geometry for our application. To help this choice we can make a contour plot for the capacitance density as a function of these two parameters. This is shown in Fig. 9 (the simulation parameters are: $\varepsilon_S = 3.15$; $\varepsilon_1 = 5$; $\lambda = 400 \mu\text{m}$).

As the analyte molecules interact with the sensitive layer, they will change its permittivity and thickness thus causing a change in capacitance. The detailed mechanisms for the change in the permittivity and thickness of the sensitive layer are related to its interaction with the analyte. These interactions can be analyzed using thermodynamics (e.g. linear solvation energy relationships (LSERs) [13]) and kinetics and are out of the scope of the present paper.

The change in the capacitance imposed by a change in the thickness of the sensitive layer can be in many cases an important factor, especially in the case of detecting low polar analytes in which the permittivity of the analyte are closed to the permittivity of the sensitive layer. In those cases we can use this model to predict the change in the sensor capacitance for a given change in the thickness of the sensitive layer.

As a practical example, let us suppose, we want a sensor of 32 pF with an area of 2 cm^2 with $\lambda = 400 \mu\text{m}$, $\varepsilon_S = 3.15$; $\varepsilon_1 = 5$. Table 1 represents five different geometries taken from Fig. 9. All of this geometries will give a nominal sen-

Table 1

Example of different geometry possibilities for an IDC-S with the same capacitance, area of implementation and spatial sensor wavelength and its influence on the relative change on the overall sensor capacitance for a change of 5% in the thickness of the sensitive layer

η	$W (\mu\text{m})$	$G (\mu\text{m})$	r	$h (\mu\text{m})$	$\Delta C/C$ (%) for $\Delta h/h = \Delta r/r = 5\%$
0.45	90	110	0.25	100	0.42
0.50	100	100	0.16	64	0.98
0.55	110	90	0.12	48	1.14
0.60	120	80	0.09	34	1.13
0.65	130	70	0.06	24	1.03

These five geometries are pointed out in Fig. 9.

$C = 32 \text{ pF}$; $A = 2 \text{ cm}^2$; $\lambda = 400 \mu\text{m}$; $\varepsilon_S = 3.15$; $\varepsilon_1 = 5$.

sor capacitance of 32 pF for a sensor with an area of 2 cm^2 , since all points are in the 16 pF/cm^2 curve (see Fig. 9). As can be seen in Table 1, the expected change in the capacitance imposed by a change in the thickness of the sensitive layer will be different for all geometries. In fact, for an expected relative change of 5% in the thickness of the sensitive layer we will have a high relative change in the sensor capacitance (1.14%) for the pair of values ($\eta = 0.55$, $r = 0.12$) which gives $h = 48 \mu\text{m}$. Notice that the high value of the capacitance change (ΔC) will occur for the same geometric parameters as the relative change ($\Delta C/C$) since we are fixing the capacitance density.

Although the method described above is appropriate to choose the sensor geometry with optimized values of the capacitance and change of capacitance, it is not efficient in practice if we want to make several IDC-S with different sensitive layers (which is our case), since every sensor will probably give a different optimum metallization ratio parameter. This means that we need to make all the manufacturing process from the beginning for each sensor. Another approach is to fix the metallization ratio parameter and choose the appropriated value of r according to the permittivity of the sensitive layer to be deposited over the electrodes.

In fact, for a particular defined sensor geometry (i.e. fixing the values of the permittivity, the sensor wavelength and the metallization ratio) the thickness of the sensitive layer can be chosen to allow the highest value of the relative change of the capacitance. This can be observed in Fig. 10, where the relative change in capacitance, for a 5% change in thickness, shows a peak for an r -value of around 0.09 (simulation parameters are: $\varepsilon_S = 3.15$; $\varepsilon_1 = 5$; $A = 2 \text{ cm}^2$; $\lambda = 400 \mu\text{m}$; $\eta = 0.5$; $\Delta h/h = \Delta r/r = 5\%$). In this particular case the thickness of the sensitive layer should be chosen to be $h = 0.09 \times 400 = 36 \mu\text{m}$. The corresponding nominal capacitance density is 14.3 pF/cm^2 .

It should be pointed out that when the relative changes in the sensor capacitance ($\Delta C/C$) are small is preferable to use ratiometric or bridge circuits to read the sensor signal. This type of circuits will cancel the nominal sensor capacitance and give a signal that is a function of the change of the sensor capacitance (ΔC). The change in capacitance

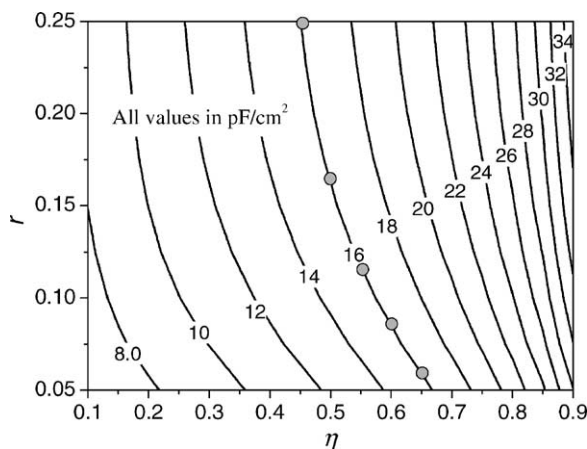


Fig. 9. Contour plot for the capacitance density (pF/cm^2) as a function of the parameter η and r for an IDC-S with $\lambda = 400 \mu\text{m}$, $\varepsilon_S = 3.15$ and $\varepsilon_1 = 5$.

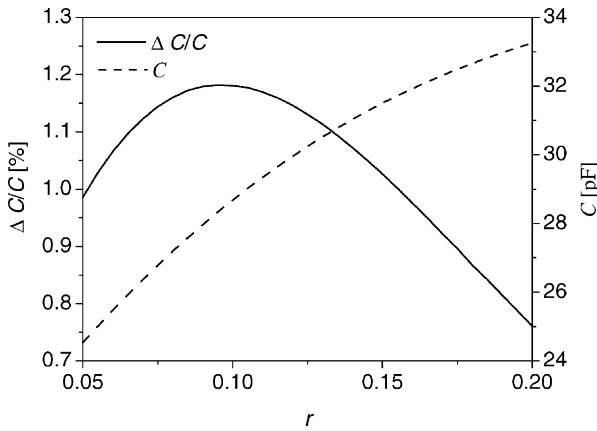


Fig. 10. Nominal sensor capacitance and relative change of the sensor capacitance as a function of the thickness of the sensitive layer. The simulation parameters are as following: $\varepsilon_S = 3.15$; $\varepsilon_1 = 5$; $A = 2 \text{ cm}^2$; $\lambda = 400 \text{ } \mu\text{m}$; $\eta = 0.5$; $\Delta h/h = \Delta r/r = 5\%$.

(ΔC) is the multiplication of the two curves in Fig. 10, and the corresponding peak is slightly moved to the right when compared with the peak for $\Delta C/C$.

When a change in the permittivity of the sensitive layer, due to the sorption of the analyte, is more a important factor for the IDC-S than a change in its thickness due to swelling, the optimum sensitive layer thickness may differ from that we found above. In this case one would prefer to have a large nominal sensor capacitance and consequently, a thick layer but not too thick that, for example, a monolayer of an adsorbed vapor on the sensitive layer would go unnoticed. In Fig. 8, one sees that a good compromise is found for $r \approx 0.2$ where the capacitance is not too far from its saturation value but still is sensitive to changes of the permittivity near the surface of the sensitive layer. In fact, we point out that Fig. 8 shows that a change in the permittivity of the sensitive layer for $r > 0.5$ will not influence significantly the overall value of the capacitance.

4. Conclusion

It has been shown that conformal mapping technique could be successfully used in conjugation with the partial capacitance technique to generate relatively simple expressions for the capacitance of planar interdigital structures. It is worth to note that all functions used in the present model (e.g. elliptic integrals of first kind) are nowadays

available in most mathematical software packages and there is no need to use approximate expressions or mathematical tables.

In this paper, new analytical expressions for the capacitance of IDC-Ss has been developed. These expressions are particular useful for our case study, i.e. a thick substrate with a dielectric sensitive layer on the top of the electrodes, although it is clear that the method described can be applied to a general multi-layered structure. Other corrections can be suitably introduced as desired (e.g. contribution from electrode thickness and fringing field corrections from the finger ends).

The model presented is in good agreement with experimental values as well as with FEM. The expressions mentioned in this paper represent only a small subset of the potential applications of this model in domains where IDCs are used. This model can be extended to find the electric field and charge distributions in this type of structures, which will be a topic of a next paper.

Finally this electrical model will be coupled to chemical, thermodynamics and kinetic models in order to give a complete description of a gas sensor. In fact, one needs to understand the interaction mechanisms between the analyte and the sensitive layer to better evaluate the capacitance changes in an IDC-S.

Appendix A. Infinite height layer

Since the above analytical method to find the capacitances C_I and C_E is not quite straightforward for the case when the height of a layer approaches infinite (i.e. Eqs. (19) and (20)), an alternative method is presented here, based again on the CM technique.

A.1. Calculation of C_I for an infinite layer ($C_{I\infty}$)

For the calculation of $C_{I\infty}$ the transformations are the following (see Fig. 11):

$$t = \frac{1}{k_{I\infty}} \sin \left(\frac{2\pi}{\lambda} z \right)$$

and

$$w = \int_0^t \frac{dw'}{\sqrt{(1-w'^2)(1-k_{I\infty}^2 w'^2)}}$$

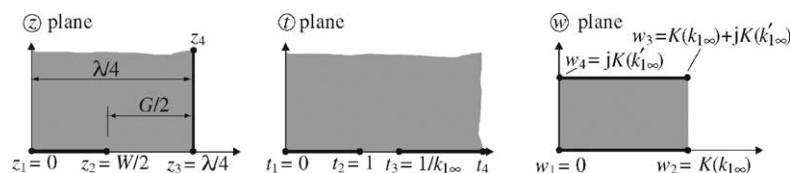


Fig. 11. Conformal transformations for the calculation of C_I for an infinite layer. The fixed equipotential lines and their transformations are marked as solid lines while the dielectric and its transformations are the shadowed regions.

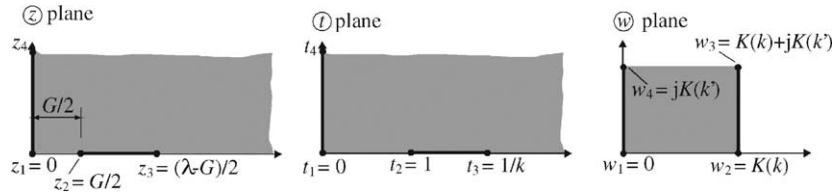


Fig. 12. Conformal transformations for the calculation of C_E for an infinite layer. The fixed equipotential lines and their transformations are marked as solid lines while the dielectric and its transformations are the shadowed regions.

with

$$k_{I\infty} = \sin\left(\frac{\pi}{2}\eta\right)$$

Thus, the capacitance is

$$C_E = \varepsilon_0 \varepsilon_r L \frac{K(k_{I\infty})}{K(k'_{I\infty})}$$

A.2. Calculation of C_E for an infinite layer ($C_{E\infty}$)

For the calculation of C_E the transformations are the following (see Fig. 12):

$$t = \frac{2}{G}z$$

and

$$w = \int_0^t \frac{dw'}{\sqrt{(1-w'^2)(1-k^2w'^2)}}$$

with

$$k = \frac{1-\eta}{1+\eta}$$

Table 2

Detailed equations needed for the calculation of C_I and C_E for a finite layer as well as for an infinite layer

	Interior electrodes	Exterior electrodes
Finite layer	$C_I = \varepsilon_0 \varepsilon_r \frac{K(k_I)}{K(k'_I)}$	$C_E = \varepsilon_0 \varepsilon_r \frac{K(k_E)}{K(k'_E)}$
	$k'_I = \sqrt{1-k_I^2}$	$k'_E = \sqrt{1-k_E^2}$
	$k_I = t_2 \sqrt{\frac{t_4^2-1}{t_4^2-t_2^2}}$	$k_E = \frac{1}{t_3} \sqrt{\frac{t_4^2-t_3^2}{t_4^2-1}}$
	$t_2 = \text{sn}(K(k)\eta, k)$	$t_3 = \cos h\left(\frac{\pi(1-\eta)}{8r}\right)$
	$t_4 = \frac{1}{k}$	$t_4 = \cos h\left(\frac{\pi(\eta+1)}{8r}\right)$
	$k = \left(\frac{v_2(0, q)}{v_3(0, q)}\right)^2$	
	$q = \exp(-4\pi r)$	
Infinite layer	$C_I = \varepsilon_0 \varepsilon_r \frac{K(k_{I\infty})}{K(k'_{I\infty})}$	$C_E = \varepsilon_0 \varepsilon_r \frac{K(k_{E\infty})}{K(k'_{E\infty})}$
	$k_{I\infty} = \sin\left(\frac{\pi}{2}\eta\right)$	$k_{E\infty} = \frac{2\sqrt{\eta}}{1+\eta}$

Thus, the capacitance is

$$C_E = \varepsilon_0 \varepsilon_r L \frac{K(k')}{K(k)} = \varepsilon_0 \varepsilon_r L \frac{K(k_{E\infty})}{K(k'_{E\infty})}$$

with

$$k_{E\infty} = \sqrt{1-k^2} = \frac{2\sqrt{\eta}}{1+\eta}$$

Appendix B. Expressions for C_I and C_E

In this appendix, Table 2 is shown with all detailed equations needed for the calculation of C_I and C_E for a finite layer as well as for an infinite layer.

Remembering that the capacitance is a function of two adimensional parameters, $\eta = W/(W+G) = 2W/\lambda$ (metalization ratio) and $r = h/\lambda$ (layer thickness to sensor wavelength ratio).

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